

1 Operators That Preserve Stability

Last time we worked through a collection of closure properties: given a real stable polynomial p , we showed that (for example) $\partial_{z_i} p$ and $p(a_1 z_1 + b_1, \dots, a_n z_n + b_n)$ for $a \in \mathbb{R}_{>0}^n, b \in \mathbb{R}^n$ are real stable or zero. But is there a systematic way to tell whether some operator preserves stability?

For $\alpha \in \mathbb{Z}_{\geq 0}^n$, recall $z^\alpha = \prod_j z_j^{\alpha_j}$ and write $\partial^\alpha = \prod_j \partial_{z_j}^{\alpha_j}$. Consider any operator $T : \mathbb{R}[z_1, \dots, z_n] \rightarrow \mathbb{R}[z_1, \dots, z_n]$ given by:

$$T = \sum_{\alpha, \beta} c_{\alpha, \beta} z^\alpha \partial^\beta, \quad c_{\alpha, \beta} \in \mathbb{R},$$

for some set of $\alpha, \beta \in \mathbb{Z}_{\geq 0}^n$. For example, the operator $T = 1 - z_2 \partial_{z_1}$ applied to a polynomial p is $p - z_2 \partial_{z_1} p$. Here we differentiate first, then multiply by z^α .

Now, associate to T the polynomial in $2n$ variables, where we create variables $w_1, \dots, w_n$:

$$p_T(z, w) = \sum_{\alpha, \beta} c_{\alpha, \beta} z^\alpha (-w)^\beta.$$

In other words, replace each differentiation ∂_{z_j} by $-w_j$.

Theorem 1.1 (Borcea-Brändén). *A differential operator $T \neq 0$ maps every real stable polynomial to a real stable polynomial or zero if and only if $p_T(z, w)$ is real stable.*

This is a pretty amazing fact: now, to check whether T works for every real stable polynomial, we only have to check stability of the single polynomial p_T . Let's try some examples:

- (1) $T = 1 - \partial_{z_1}$ has $p_T = 1 + w_1$, so it preserves stability.
- (2) $T = 1 + z_2 \partial_{z_1}$ has $p_T = 1 - z_2 w_1$, which is real stable by our example $1 - xy$. So this operator preserves stability too.
- (3) $T = 1 - \partial_{z_1} \partial_{z_2}$ has $p_T = 1 - w_1 w_2$, again real stable.
- (4) In contrast, $1 + \partial_{z_1} \partial_{z_2}$ does not preserve stability: it takes the stable polynomial $z_1 z_2$ to $z_1 z_2 + 1$, which vanishes at (i, i) . Or, we can see that $p_T(z, w) = 1 + w_1 w_2$ is not stable (same polynomial).

1.1 Multi-Affine Part (MAP)

Consider the operation which only takes the multi-affine monomials of a polynomial. In other words, delete every monomial in which some variable has exponent at least 2. Call the resulting polynomial $\text{MAP}(p)$. For example:

$$\text{MAP}(1 + 2x + xy + x^2 y + z^3) = 1 + 2x + xy.$$

At first, it looks difficult to reason about whether this is a stability preserver. But using [Theorem 1.1](#) it is easy to prove that it is.

Lemma 1.2. *If $p \in \mathbb{R}[z_1, \dots, z_n]$ is real stable, then $\text{MAP}(p)$ is real stable or zero.*

Proof. Introduce new variables $z'_1, \dots, z'_n$ which will take the place of $z_1, \dots, z_n$. We can write:

$$\text{MAP}(p) = \left[\prod_{j=1}^n (1 + z'_j \partial_{z_j}) \right] p(z) \Big|_{z_1=\dots=z_n=0}.$$

where $p(z)|_{z_1=\dots=z_n=0}$ means to specialize 0 in place of all $z_1, \dots, z_n$. To check this identity, consider any monomial. For each variable z_i , if it has degree 0 in the monomial, it "survives" by choosing the identity term. If it has degree 1, it survives as z'_i by choosing $z'_i \partial_{z_i}$. If it has degree at least 2, evaluation at $z_i = 0$ kills the term.

Each operator $1 + z'_j \partial_{z_j}$ preserves stability, by the symbol $1 - z'_j w_j$. So their product is real stable by closure properties. Then, specializing $z_i = 0$ for all i preserves stability. $\square$

2 Another Proof of Heilmann-Lieb

MAP turns out to give us an alternate proof of Heilmann-Lieb. Consider:

$$q_G(z) = \prod_{\{u,v\} \in E} (1 - w_{\{u,v\}} z_u z_v).$$

Every polynomial in the product is real stable, so q_G is real stable.

Each monomial corresponds to a set of selected edges, and every selected edge contributes its weight, a minus sign, and a variable for each endpoint. So, each monomial only survives MAP when that set forms a matching, giving:

$$M_G(z) = \text{MAP}(q_G)(z) = \sum_{M \in \mathcal{M}} (-1)^{|M|} \left(\prod_{e \in M} w_e \right) \prod_{v \in V(M)} z_v,$$

where $V(M)$ is the set of vertices covered by M . M_G is real stable because MAP preserves stability (and not zero, because of the constant monomial 1). Often, M_G is called the matching polynomial instead of the univariate p_G from Lecture 2.

But, we can recover stability of the p_G from Lecture 2. Recall:

$$p_G(t) = \sum_k m_k t^k, \quad m_k = \sum_{M \in \mathcal{M}: |M|=k} \prod_{e \in M} w_e.$$

Set every variable of M_G equal to t . Since a matching with k edges covers $2k$ vertices, we get:

$$M_G(t, \dots, t) = \sum_k (-1)^k m_k t^{2k} = p_G(-t^2).$$

This is non-zero and real rooted by diagonalization, so it means $p_G(-t^2)$ is real rooted. This implies p_G is real rooted: suppose p_G had a nonreal root ρ . Take t so that $-t^2 = \rho$. Note t must be nonreal: if it was real, then ρ would be real too. But then $p_G(-t^2) = p_G(\rho) = 0$, contradiction.

3 Approximating the Permanent and Counting Perfect Matchings

This brings us to our first algorithmic result, which will also be related to matchings. Let $M \in \mathbb{R}_{\geq 0}^{n \times n}$. The permanent of M is:

$$\text{per}(M) = \sum_{\sigma \in S_n} \prod_{i=1}^n M_{i,\sigma(i)}$$

A natural question is whether given a non-negative matrix, we can compute the permanent in polynomial time. Unfortunately, it is #P-hard,¹ but a randomized $1 + \epsilon$ approximation was designed by Jerrum, Sinclair and Vigoda which runs in polynomial time for any $\epsilon > 0$. A major open question in computer science is whether there exists a *deterministic* $1 + \epsilon$ approximation for this problem. Notice that if $M \in \{0, 1\}^{n \times n}$ is the adjacency matrix of a bipartite graph, then the permanent exactly counts the number of perfect matchings.

Here, we will show a deterministic e^n approximation for the permanent on non-negative matrices. This is close to the best deterministic approximation, which is slightly below $\sqrt{2}^n$. Our approach will be as follows. First, we will show that $\text{per}(M)$ is the coefficient of a monomial of a real stable polynomial p . Then, we will show a way to approximate the coefficient of that monomial of p using a convex program that can be solved in polynomial time.

First, let's find p . Here's the idea: associate with each column of M a variable z_i so that we have $p(z_1, \dots, z_n)$, and consider choosing one element from each row. Then, the coefficient of $z_1 z_2 \dots z_n$ will exactly be the permanent. In particular, given M , let:

$$p(z_1, \dots, z_n) = \prod_{i=1}^n \left(\sum_{j=1}^n M_{ij} z_j \right)$$

So long as no row is 0 (in which case the permanent would be zero), this polynomial is the product of real stable polynomials, so it itself is real stable. Every monomial will have one term from each row, and the monomial $z_1 z_2 \dots z_n$ will also have one term from each column. Therefore:

$$\text{per}(M) = \partial_{z_1} \partial_{z_2} \dots \partial_{z_n} p \Big|_{z=0}$$

This just means take derivatives, and then evaluate at 0. We can equivalently write $(\partial_{z_1} \partial_{z_2} \dots \partial_{z_n} p)(0)$. So, it extracts the coefficient of $z_1 z_2 \dots z_n$. Notice that while we can easily write the polynomial in this product form in polynomial time, it's no easy feat to actually extract the coefficients of the monomials (and as we said, it's actually #P-hard).

3.1 Capacity

The *capacity* of a polynomial at a point $\beta \in \mathbb{R}_{\geq 0}^n$ is the following:

$$\text{cap}_{\beta}(p) = \inf_{z > 0} \frac{p(z)}{z^{\beta}}$$

¹Recall #P-hard problems are "harder" than NP-hard problems. Also, note that if the matrix can have both positive and negative entries, it is #P-hard to even approximate the permanent.

in this lecture we will be interested in $\text{cap}_1(p) = \inf_{z>0} \frac{p(z)}{z_1 \dots z_n}$. Often we will just write $\text{cap}(p)$ to mean $\text{cap}_1(p)$, the capacity at the all 1s vector.

The capacity at a point is particularly interesting because it models the Newton polytope, the convex hull of the support of p .

Definition 3.1 (Newton Polytope). Given a polynomial $p \in \mathbb{R}_{\geq 0}[z_1, \dots, z_n]$, let $p(z_1, \dots, z_n) = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^n} c_\alpha z^\alpha$. The Newton polytope of p is the convex hull of all α for which $c_\alpha > 0$. If $C_p \subseteq \mathbb{Z}_{\geq 0}^n$ is the set of α for which $c_\alpha > 0$, then:

$$\text{NewtonPolytope}(p) = \{\beta \in \mathbb{R}^n : \beta = \sum_{\alpha \in C_p} \lambda_\alpha \alpha \text{ for some } \lambda \in \mathbb{R}_{\geq 0}^n, \|\lambda\|_1 = 1\}$$

Capacity gives us a way to test for membership in the Newton polytope for any polynomial with non-negative coefficients:

Lemma 3.2. For any $p \in \mathbb{R}_{\geq 0}[z_1, \dots, z_n]$, $\beta \in \text{NewtonPolytope}(p)$ if and only if $\text{cap}_\beta(p) > 0$.

To prove this, we will need the weighted AM-GM inequality, which we recall now:

Lemma 3.3 (Weighted AM-GM). Let $\lambda \in \mathbb{R}_{\geq 0}^n$, $\|\lambda\|_1 > 0$. Then for any $x \in \mathbb{R}_{\geq 0}^n$:

$$\frac{\sum_{i=1}^n \lambda_i x_i}{\|\lambda\|_1} \geq \left(\prod_{i=1}^n x_i^{\lambda_i} \right)^{1/\|\lambda\|_1}$$

when $\|\lambda\|_1 = 1$, this has the simpler statement:

$$\sum_{i=1}^n \lambda_i x_i \geq \prod_{i=1}^n x_i^{\lambda_i}$$

Note that if $\lambda_i = \frac{1}{n}$ for all i , this recovers AM-GM.

Now let's prove [Lemma 3.2](#).

Proof. We will prove that if β is in the Newton polytope, then the capacity is positive, and leave the other direction as an exercise. Since β is in the polytope, we can write it as a convex combination of a set of vertices $S \subseteq \mathbb{Z}_{\geq 0}^n$ of the polytope.² So for some non-negative λ , $\|\lambda\|_1 = 1$:

$$\beta = \sum_{\alpha \in S} \lambda_\alpha \alpha$$

where WLOG $\lambda_\alpha > 0$ for all $\alpha \in S$. Now:

$$p(z) = \sum_{\alpha} c_\alpha z^\alpha \geq \sum_{\alpha \in S} c_\alpha z^\alpha = \sum_{\alpha \in S} \lambda_\alpha \frac{c_\alpha z^\alpha}{\lambda_\alpha} \geq \prod_{\alpha \in S} \left(\frac{c_\alpha z^\alpha}{\lambda_\alpha} \right)^{\lambda_\alpha} = z^\beta \prod_{\alpha \in S} \left(\frac{c_\alpha}{\lambda_\alpha} \right)^{\lambda_\alpha}$$

where in the last expression we used that $\beta = \sum_{\alpha \in S} \lambda_\alpha \alpha$. Therefore:

$$\text{cap}_\beta(p) = \inf_{z>0} \frac{p(z)}{z^\beta} \geq \prod_{\alpha \in S} \left(\frac{c_\alpha}{\lambda_\alpha} \right)^{\lambda_\alpha} > 0$$

where here we use that all coefficients c_α are positive and $\lambda_\alpha > 0$ for all $\alpha \in S$. □

²By Carathéodory's theorem, we can choose S so that $|S| \leq n + 1$.

3.2 Bound on the Permanent

In the following lecture, we will prove this theorem of Gurvits:

Theorem 3.4 (Gurvits). *Let $p \in \mathbb{R}_{\geq 0}[z_1, \dots, z_n]$ be real stable. Then:*

$$e^{-n} \text{cap}_1(p) \leq \partial_{z_1} \dots \partial_{z_n} p|_{z=0} \leq \text{cap}_1(p)$$

We will then argue that $\text{cap}_1(p)$ can be computed in polynomial time (up to small error) using a convex program, giving a deterministic e^n approximation to the permanent. We will also show how this leads to Gurvits' proof of the Van der Waerden conjecture:

Theorem 3.5 (Van der Waerden Conjecture). *Let $M \in \mathbb{R}_{\geq 0}^{n \times n}$ be a doubly stochastic matrix, i.e., all row sums and column sums are 1. Then:*

$$\text{per}(M) \geq e^{-n}$$

in fact, one can obtain the slightly stronger $\text{per}(M) \geq \frac{n!}{n^n}$ with slightly more careful accounting, the original form of the conjecture.

To show this, we will simply prove that the capacity of the permanent polynomial at all 1s is at least 1, which with [Theorem 3.4](#) gives the theorem.